

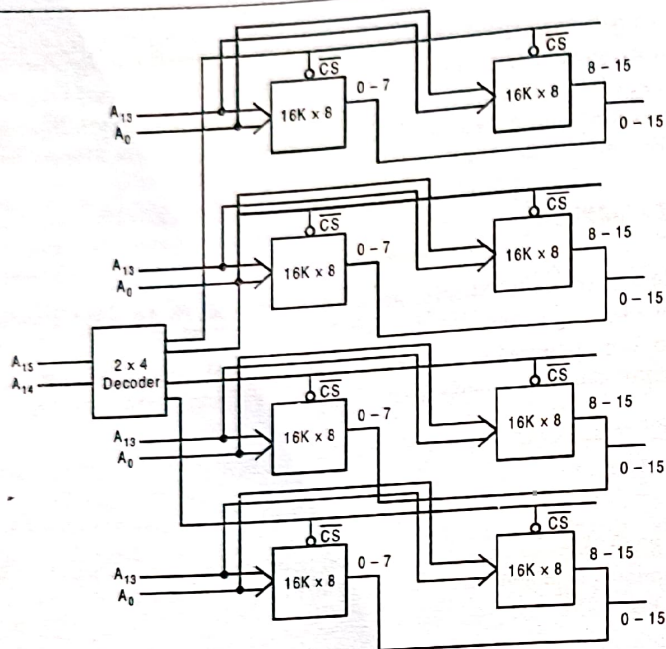


Fig. 6.26.

Example 6.5. (a) How many 128×8 RAM chips are needed to provide a memory capacity of 2048 bytes?

(b) How many lines of the address bus must be used to access 2048 bytes of memory? How many of these lines will be common to all chips?

(c) How many lines must be decoded for chip select? Specify the size of the decoder.

Solution. (a) Number of chips required

$$= \frac{2048 \times 8}{128 \times 8} = 16 \text{ chips}$$

(b) Since total memory is of 2048 i.e. 2^{11} bytes. Hence, 11 lines of the address bus is required to access 2048 bytes.

Also, RAM chip is of $128 = 2^7$ bytes. So, 7 lines out of 11 will be common to all chips.

(c) 4 lines must be decoded for selecting 16 chips. Thus, the size of decoder required is 4×16 .

Example 6.6. A digital computer has a memory unit of $64K \times 16$ and a cache memory of 1K words. The cache uses direct mapping with a block size of four words.

(a) How many bits are there in the tag, index, block and words fields of the address format?

(b) How many bits are there in each word of cache, and how are they divided into functions? Include a valid bit.

(c) How many blocks can the cache accommodate?

Solution. Given :

Main memory is of size $64K \times 16$ words

$\therefore$ 16 bits is required to address each word of main memory. Also each word is of 16 bits.

$$\Rightarrow M = 16$$

Cache memory is of size 1 K words

$\therefore$ 10 bits is required to address each word of cache.

$$\Rightarrow N = 10$$

Direct mapping is used and block size is of four words.

(a) Since cache memory address is of 10 bits

$\therefore$ Index will be of 10 bits.

$$\text{Tag} = M - N$$

$$= 16 - 10$$

$$= 6 \text{ bits}$$

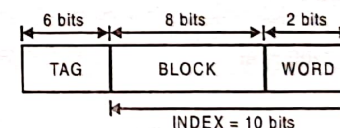
Cache consists of 1 K words and there are 4 words in 1 block

$$\begin{aligned} \therefore \text{Number of blocks} &= \frac{1K}{4} = \frac{1024}{4} \\ &= 256 \text{ blocks} \\ &= 2^8 \text{ blocks} \end{aligned}$$

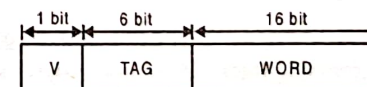
Thus, block will be of 8 bits.

Also, 4 words per block means 2^2 words per block, therefore word will be of 2 bits.

Thus, address format is



(b) Each word of cache consists of 16-bit word from main memory along with the tag. A valid bit will also be included into each cache word. Hence, the word format of cache is



$\therefore$ Each word of cache is of 23 bits.

$$(c) \text{ Number of blocks} = \frac{1K}{4} = \frac{1024}{4} = 256$$

$\therefore$ The cache can accommodate 256 blocks of 4 words each.

Example 6.7. A block-set-associative cache consists of a total of 64 blocks divided

into 4 blocks sets. The main memory contains 4096 blocks each consisting of 128 words.

- (i) How many bits are there in main memory address ?
 (ii) How many bits are there in each of the TAG, SET and WORD fields.

Solution.

$$\begin{aligned} \text{(i) Main memory size} &= 4096 \text{ blocks} \\ &= 4096 \times 128 \text{ words} \\ &= 2^{12} \times 2^7 \text{ words} \\ &= 2^{19} \text{ words} \end{aligned}$$

Thus, main memory address will consists of 19 bits.

$$\begin{aligned} \text{(ii) Main memory size} &= 4096 \text{ blocks.} \\ \Rightarrow M &= 2^{12} \text{ blocks and } m = 12 \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Cache memory size} &= 64 \text{ blocks} \\ \Rightarrow N &= 2^6 \text{ blocks and } n = 6. \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Number of cache sets} &= \frac{64}{4} = 16 \\ &= 2^4 \end{aligned}$$

$$\text{Set size} = 4 \text{ blocks}$$

$$S = 2^2 \text{ and } s = 2.$$

$$\Rightarrow \text{Tag} = m - n + s = 12 - 6 + 2 = 8 \text{ bits.}$$

$$\text{Set} = n - s = 6 - 2 = 4 \text{ bits}$$

$$\text{Word} = 128 = 2^7 \Rightarrow 7 \text{ bits.}$$

Example 6.8. A certain magnetic disk has the following specification :

- Number of recording = 10
- Number of tracks/surface = 256
- Number of sectors/track = 32
- Number of bytes/sector = 128
- Disk rotation speed = 2400 rpm

Calculate the following :

(i) Disk capacity

(ii) Transfer rate

(iii) Latency time.

Solution.

$$\begin{aligned} \text{(i) Disk capacity} &= \text{Number of surfaces} \times \text{Number of tracks in each surface} \times \text{Number} \\ &\quad \text{of sectors in a track} \times \text{Number of bytes in a sector.} \end{aligned}$$

$$= 10 \times 256 \times 32 \times 128$$

$$= 10 \times 2^8 \times 2^5 \times 2^7$$

$$= 10 \times 2^{20}$$

$$= 10 \text{ MB}$$

$$\begin{aligned} \text{(ii) Capacity of one track} &= \text{Number of sectors in a track} \times \text{Number of bytes in a sector} \\ &= 32 \times 128 = 2^5 \times 2^7 = 2^{12} \text{ bytes} \end{aligned}$$

Also given, disk rotation speed = 2400 rpm

$$\therefore 2400 \text{ rotation in 1 min i.e. 60 sec.}$$

$$\Rightarrow 1 \text{ rotation in } \frac{60}{2400} \text{ sec} = \frac{1}{40} \text{ sec.}$$

$$\text{In } \frac{1}{40} \text{ second, data transfer} = 2^{12} \text{ bytes.}$$

$$\begin{aligned} \therefore \text{In 1 sec., data transfer} &= 2^{12} \times 40 \text{ bytes} \\ &= 40 \times 4 \times 2^{10} \text{ bytes} \\ &= 160 \text{ K bytes} \end{aligned}$$

$$\therefore \text{Transfer rate} = 160 \text{ KB/sec}$$

$$\text{Latency time} = \frac{1}{2} \text{ (time in one notation)}$$

$$= \frac{1}{2} \times \frac{1}{40}$$

$$= \frac{1}{80} \text{ seconds.}$$

Example 6.9. An address space is specified by 24 bits and the corresponding memory space by 16 bits.

(a) How many words are there in the address space ?

(b) How many words are there in the memory space ?

(c) If a page consists of 2K words, how many pages and blocks are there in the system ?

Solution.

(a) Address space consists of 24 bits.

$$\therefore 2^{24} = 16 \text{ M words are there in the address space.}$$

(b) Memory space consists of 16 bits

$$\therefore 2^{16} = 64 \text{ K words are there in the memory space.}$$

(c) Page consists of 2K words

$$\begin{aligned} \therefore \text{Number of pages} &= \frac{16\text{M}}{2\text{K}} \\ &= 8 \text{ K pages} \end{aligned}$$

$$\text{and Number of blocks} = \frac{64\text{K}}{2\text{K}} = 32 \text{ blocks}$$

Hence, the system consists of 8K pages and 32 blocks.

Example 6.10. A computer employs RAM chips of 256×8 and ROM chips of 1024×8 . The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units, each with four registers. A memory-mapped I/O configuration is used. The two highest-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers.

(a) How many RAM and ROM chips are needed ?

(b) Draw a memory-address map for the system.

(c) Give the address range in hexadecimal for RAM, ROM and interface.